

Numerical Solution of Modified Shock Equations

Project II

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Abstract

The structure of a cosmic-ray modified shock was investigated using Chevalier's self-similar formulation. The governing dimensionless Euler equations were reduced to a system of coupled ordinary differential equations and solved numerically using the sixth-order Runge–Kutta method of Luther (1968). Profiles of the dimensionless density, gas pressure, cosmic-ray pressure, and velocity were obtained for a strongly modified shock with $\beta = 0.5$, $\gamma_g = 5/3$, and $\gamma_{CR} = 4/3$. The compression ratio, similarity parameter B , and energy fractions were calculated. The results show that the cosmic-ray component contains the largest fraction of the explosion energy, demonstrating the significant influence of cosmic rays on the shock structure.

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1 Introduction

Supernova remnants (SNRs) are among the primary sites of strong shock waves in astrophysics. As a supernova explosion expands into the surrounding interstellar medium, a significant fraction of its energy is converted into thermal energy, kinetic motion, and accelerated particles. The evolution of the remnant is therefore closely linked to the physics of shock propagation.

The classical description of an adiabatic blast wave is provided by the Sedov–Taylor solution [1, 2], which assumes that the explosion energy is conserved and that the shocked medium behaves as a single ideal fluid. Under these assumptions, the hydrodynamic variables become self-similar functions of the dimensionless coordinate ($\xi = r/R$), where (R) is the shock radius.

Observations, however, suggest that supernova remnant shocks efficiently accelerate cosmic rays. The pressure associated with these relativistic particles can become dynamically important and modify the internal structure of the shock. As a result, the standard single-fluid description is no longer sufficient. To address this problem, Chevalier (1983) extended the self-similar blast-wave solution by including a separate cosmic-ray pressure component in addition to the thermal gas pressure [3].

The aim of this project is to investigate the structure of a strongly modified shock using Chevalier’s self-similar formulation. The dimensionless Euler equations are reduced to a system of coupled ordinary differential equations and solved numerically using the sixth-order Runge–Kutta method of Luther (1968) [4]. The resulting density, gas pressure, cosmic-ray pressure, and velocity profiles are analysed, and the compression ratio, similarity parameter (B), and energy fractions are calculated.

2 Theoretical Background

2.1 Modified Shocks and Cosmic-Ray Pressure

The standard hydrodynamic description of a supernova remnant assumes that the shocked material behaves as a single ideal gas. In this case, the dynamics of the blast wave are governed by the conservation of mass, momentum, and energy, leading to the well-known Sedov–Taylor solution. However, observations of supernova remnants indicate that their shocks efficiently accelerate charged particles to relativistic energies. These relativistic particles, commonly referred to as cosmic rays, contribute a non-negligible pressure component behind the shock front.

Unlike thermal gas particles, cosmic rays possess a softer equation of state and are characterized by a different adiabatic index. Consequently, the shocked medium can no longer be described by a single fluid. Instead, it is treated as a two-component system consisting of a thermal gas and a cosmic-ray fluid. The additional pressure associated with cosmic rays modifies the internal structure of the remnant and alters the distribution of energy within the expanding shock.

To account for these effects, Chevalier (1983) extended the classical self-similar blast-wave solution by including a separate cosmic-ray pressure component [3]. The resulting model provides

a framework for studying strongly modified shocks in which cosmic rays play a significant dynamical role.

2.2 Dimensionless Variables

Following Chevalier's formulation [3], the physical variables are expressed in terms of dimensionless quantities. The similarity coordinate is defined as

$$\xi = \frac{r}{R},$$

where r is the radial coordinate and R is the shock radius. The hydrodynamic variables are written in dimensionless form as

$$\rho = \frac{\gamma + 1}{\gamma - 1} \rho_0 F(\xi), \tag{1}$$

$$P_g = \frac{2\beta}{\gamma + 1} \rho_0 \dot{R}^2 G(\xi), \tag{2}$$

$$P_{CR} = \frac{2(1 - \beta)}{\gamma + 1} \rho_0 \dot{R}^2 H(\xi), \tag{3}$$

$$v = \frac{2}{\gamma + 1} \dot{R} U(\xi), \tag{4}$$

where $F(\xi)$, $G(\xi)$, $H(\xi)$, and $U(\xi)$ represent the dimensionless density, gas pressure, cosmic-ray pressure, and velocity profiles, respectively. The parameter β denotes the fraction of post-shock pressure associated with the thermal gas, while $1 - \beta$ represents the fraction associated with cosmic rays.

The introduction of dimensionless variables transforms the problem from a time-dependent partial differential equation system into a self-similar problem described solely by the coordinate ξ . This significantly simplifies the analysis while preserving the essential physics of the shock evolution.

2.3 Dimensionless Euler Equations

The evolution of the modified shock is governed by the Euler equations for mass conservation, momentum conservation, and adiabatic evolution of both pressure components. In dimensional form, these equations are

$$\frac{\partial \rho}{\partial t} + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho v) = 0, \quad (5)$$

$$\rho \frac{Dv}{Dt} = -\frac{\partial P_g}{\partial r} - \frac{\partial P_{CR}}{\partial r}, \quad (6)$$

$$\frac{D}{Dt} (P_g \rho^{-\gamma_g}) = 0, \quad (7)$$

$$\frac{D}{Dt} (P_{CR} \rho^{-\gamma_{CR}}) = 0. \quad (8)$$

After substituting the self-similar variables and following the procedure outlined by Chevalier, the governing equations become

$$\left(\frac{U}{\xi} - \frac{\gamma+1}{2} \right) \xi \frac{F'}{F} + U' + \frac{2U}{\xi} = 0, \quad (9)$$

$$\xi \left[\left(\frac{2}{\gamma+1} \frac{U}{\xi} - 1 \right) U' - \frac{3}{2} \frac{U}{\xi} \right] + \frac{\gamma-1}{\gamma+1} \frac{1}{F} [\beta G' + (1-\beta)H'] = 0, \quad (10)$$

$$GF^{1-\gamma_g} \left(\frac{U}{\xi} - \frac{\gamma+1}{2} \right) = \frac{1-\gamma}{2\xi^3}, \quad (11)$$

$$HF^{1-\gamma_{CR}} \left(\frac{U}{\xi} - \frac{\gamma+1}{2} \right) = \frac{1-\gamma}{2\xi^3}. \quad (12)$$

These equations describe the complete self-similar structure of a cosmic-ray modified shock. The first equation represents mass conservation, the second represents momentum conservation, while the remaining two equations describe the adiabatic evolution of the thermal gas and cosmic-ray components. In the following section, these equations are manipulated to obtain explicit expressions for $dU/d\xi$ and $dF/d\xi$, which are subsequently solved numerically.

3 Derivation of the Governing Equations

The dimensionless Euler equations presented in the previous section form a coupled system involving the variables F , G , H , and U . For numerical integration, it is convenient to reduce the system to two first-order ordinary differential equations involving only the density and velocity profiles. This can be achieved by eliminating the pressure variables G and H using the adiabatic relations [5].

3.1 Expressions for the Pressure Components

From the adiabatic equations,

$$GF^{1-\gamma_g} \left(\frac{U}{\xi} - \frac{\gamma+1}{2} \right) = \frac{1-\gamma}{2\xi^3}, \quad (13)$$

$$HF^{1-\gamma_{CR}} \left(\frac{U}{\xi} - \frac{\gamma+1}{2} \right) = \frac{1-\gamma}{2\xi^3}, \quad (14)$$

the gas and cosmic-ray pressure profiles can be written explicitly as

$$G = \frac{(1-\gamma)}{2\xi^3} \frac{F^{\gamma_g-1}}{\left(\frac{U}{\xi} - \frac{\gamma+1}{2} \right)}, \quad (15)$$

and

$$H = \frac{(1-\gamma)}{2\xi^3} \frac{F^{\gamma_{CR}-1}}{\left(\frac{U}{\xi} - \frac{\gamma+1}{2} \right)}. \quad (16)$$

These relations express both pressure components solely in terms of the density and velocity profiles.

3.2 Reduction of the System

The continuity equation,

$$\left(\frac{U}{\xi} - \frac{\gamma+1}{2} \right) \xi \frac{F'}{F} + U' + \frac{2U}{\xi} = 0, \quad (17)$$

provides a relation between the derivatives of F and U .

Substituting the pressure expressions into the momentum equation and differentiating the pressure terms with respect to ξ yields a coupled system containing only F , U , F' , and U' .

After algebraic manipulation and elimination of the pressure derivatives, the governing equations can be written in explicit form as

$$\frac{dU}{d\xi} = f(\xi, F, U), \quad (18)$$

and

$$\frac{dF}{d\xi} = g(\xi, F, U), \quad (19)$$

where f and g are nonlinear functions obtained from the continuity and momentum equations.

3.3 Final Form of the Differential Equations

The explicit expression for the velocity derivative is

$$\begin{aligned} \frac{dU}{d\xi} = \frac{1}{\xi} & \left[\frac{(\gamma - 1)^2 \left\{ \beta F^{\gamma_g - 2} \left[\frac{3}{2}(\gamma + 1)\xi - 2\gamma_g U \right] + (1 - \beta) F^{\gamma_{CR} - 2} \left[\frac{3}{2}(\gamma + 1)\xi - 2\gamma_{CR} U \right] \right\}}{(\gamma - 1)^2 [\beta \gamma_g F^{\gamma_g - 2} + (1 - \beta) \gamma_{CR} F^{\gamma_{CR} - 2}] + 4\xi^2 \left(U - \frac{\gamma + 1}{2} \xi \right)^3} \right. \\ & \left. + \frac{3(\gamma + 1)\xi^3 U \left(U - \frac{\gamma + 1}{2} \xi \right)^2}{(\gamma - 1)^2 [\beta \gamma_g F^{\gamma_g - 2} + (1 - \beta) \gamma_{CR} F^{\gamma_{CR} - 2}] + 4\xi^2 \left(U - \frac{\gamma + 1}{2} \xi \right)^3} \right]. \end{aligned} \quad (20)$$

Similarly, the density derivative becomes

$$\begin{aligned} \frac{dF}{d\xi} = -\frac{F}{U - \frac{\gamma + 1}{2} \xi} & \left[\frac{\frac{3}{2}(\gamma + 1)(\gamma - 1)^2 [\beta F^{\gamma_g - 2} + (1 - \beta) F^{\gamma_{CR} - 2}]}{(\gamma - 1)^2 [\beta \gamma_g F^{\gamma_g - 2} + (1 - \beta) \gamma_{CR} F^{\gamma_{CR} - 2}] + 4\xi^2 \left(U - \frac{\gamma + 1}{2} \xi \right)^3} \right. \\ & \left. + \frac{\xi U [8U - (\gamma + 1)\xi] \left(U - \frac{\gamma + 1}{2} \xi \right)^2}{(\gamma - 1)^2 [\beta \gamma_g F^{\gamma_g - 2} + (1 - \beta) \gamma_{CR} F^{\gamma_{CR} - 2}] + 4\xi^2 \left(U - \frac{\gamma + 1}{2} \xi \right)^3} \right]. \end{aligned} \quad (21)$$

Equations (20) and (21) constitute a closed system of first-order ordinary differential equations. These equations form the basis of the numerical calculations presented in the following section.

4 Numerical Method

4.1 Boundary Conditions

The self-similar solution is defined within the interval $0 \leq \xi \leq 1$, where $\xi = 1$ corresponds to the shock front. Following Chevalier's formulation, the dimensionless variables satisfy the shock boundary conditions

$$F(1) = 1, \quad (22)$$

$$G(1) = 1, \quad (23)$$

$$H(1) = 1, \quad (24)$$

$$U(1) = 1. \quad (25)$$

These conditions correspond to the normalized post-shock values of density, gas pressure, cosmic-ray pressure, and velocity, respectively. Since the governing equations become singular at the origin, the numerical integration was initiated at the shock front and performed inward toward the centre of the remnant.

For the present calculations, the parameters were fixed at

$$\beta = 0.5, \quad (26)$$

$$\gamma_g = \frac{5}{3}, \quad (27)$$

$$\gamma_{CR} = \frac{4}{3}, \quad (28)$$

which correspond to a strongly modified shock.

The effective adiabatic index is

$$\gamma = \frac{5 + 3(1 - \beta)}{3(2 - \beta)} = \frac{13}{9} \approx 1.4444. \quad (29)$$

The corresponding total compression ratio is

$$R = \frac{\gamma + 1}{\gamma - 1} = 5.5. \quad (30)$$

4.2 Luther Sixth-Order Runge–Kutta Scheme

The coupled differential equations derived in the previous section were solved numerically using the explicit sixth-order Runge–Kutta method developed by Luther (1968) [4]. This method was selected because it was explicitly required in the project specification and provides higher-order accuracy than the classical fourth-order Runge–Kutta method.

For a general system

$$\frac{d\mathbf{y}}{d\xi} = \mathbf{f}(\xi, \mathbf{y}), \quad (31)$$

the numerical solution is advanced according to

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{9k_1 + 64k_3 + 49k_5 + 49k_6 + 9k_7}{180}, \quad (32)$$

where $k_1, \dots, k_7$ are intermediate stage evaluations defined by the Luther scheme.

In the present work the solution vector was defined as

$$\mathbf{y} = \begin{bmatrix} F \\ U \end{bmatrix}, \quad (33)$$

while the right-hand side was constructed from Equations (21) and (20).

4.3 Numerical Integration Procedure

The integration was performed from the shock front ($\xi = 1$) toward the centre of the remnant. A constant step size

$$h = -10^{-4} \quad (34)$$

was adopted, where the negative sign indicates inward integration.

At each integration step, the density and velocity profiles were updated using the Luther sixth-order Runge–Kutta method. Once the numerical solutions for $F(\xi)$ and $U(\xi)$ had been obtained, the pressure profiles $G(\xi)$ and $H(\xi)$ were reconstructed from the adiabatic relations.

The resulting dimensionless profiles were subsequently used to calculate the similarity parameter B and the fractional contributions of kinetic energy, thermal gas energy, and cosmic-ray energy through numerical evaluation of the energy integrals.

5 Results and Discussion

5.1 Dimensionless Shock Profiles

The coupled ordinary differential equations were integrated numerically using the sixth-order Runge–Kutta method of Luther (1968). The resulting dimensionless density, gas pressure, cosmic-ray pressure, and velocity profiles are shown in Figure 1.

The density profile increases monotonically from the centre of the remnant toward the shock front. Near the centre, the density approaches zero, indicating a highly rarefied interior region. As the shock front is approached, the density rises rapidly and reaches its normalized post-shock value at $\xi = 1$.

A similar trend is observed for the gas pressure profile. The thermal pressure remains small throughout much of the remnant interior and increases sharply near the shock front. This behaviour reflects the fact that gas heating is strongest immediately behind the shock, where the kinetic energy of the expanding blast wave is converted into thermal energy.

The cosmic-ray pressure exhibits a markedly different behaviour. Unlike the gas pressure, the cosmic-ray pressure remains significant throughout the entire remnant and does not vanish near the centre. This result demonstrates that relativistic particles retain a substantial fraction of the explosion energy and are distributed over a much larger volume than the thermal gas. Consequently, cosmic rays contribute significantly to the internal dynamics of the remnant.

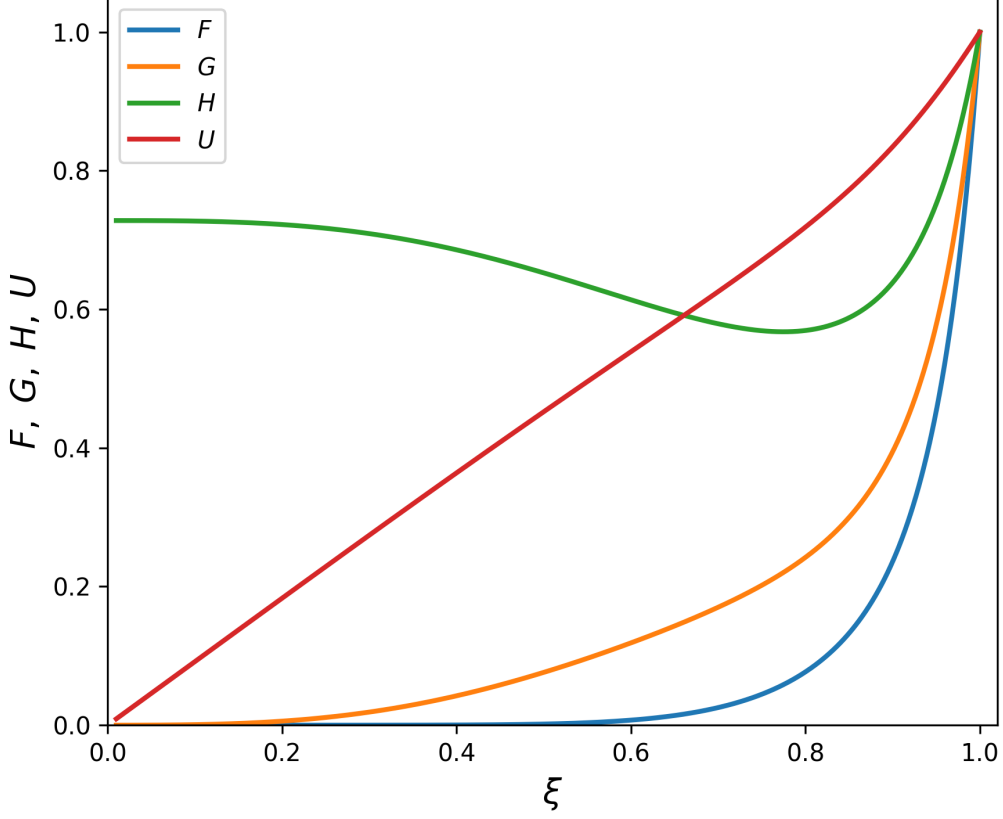


Figure 1: Numerical solution of the modified shock equations obtained using the sixth-order Runge–Kutta method for $\beta = 0.5$, $\gamma_g = 5/3$, and $\gamma_{CR} = 4/3$. The dimensionless density $F(\xi)$, gas pressure $G(\xi)$, cosmic-ray pressure $H(\xi)$, and velocity $U(\xi)$ are plotted as functions of the similarity coordinate $\xi = r/R$.

The velocity profile increases approximately linearly with the similarity coordinate. The velocity approaches zero near the centre and reaches its maximum value at the shock front. This behaviour is consistent with the expected self-similar expansion of an adiabatic blast wave.

Overall, the obtained profiles reproduce the characteristic structure of a strongly modified shock and are in qualitative agreement with the reference solution presented in the project description.

5.2 Compression Ratio and Similarity Parameter

For the adopted parameters

$$\beta = 0.5, \quad \gamma_g = \frac{5}{3}, \quad \gamma_{CR} = \frac{4}{3},$$

the effective adiabatic index was found to be

$$\gamma = 1.4444.$$

This yields a total compression ratio

$$R = \frac{\gamma + 1}{\gamma - 1} = 5.5.$$

The similarity parameter obtained from numerical evaluation of the energy integral is

$$B = 1.0342.$$

The value of B determines the normalization of the shock radius in the self-similar solution

$$R(t) = B \left(\frac{E_0}{\rho_0} \right)^{1/5} t^{2/5}.$$

The calculated value is of the expected order of magnitude for a modified Sedov-type blast wave and confirms the consistency of the numerical solution.

5.3 Energy Distribution

The kinetic, thermal gas, and cosmic-ray contributions to the total explosion energy were calculated using the energy integrals derived from the self-similar solution. The numerical results are summarized in Table 1.

Table 1: Numerical results obtained for the modified shock solution.

Quantity	Value
Effective adiabatic index γ	1.4444
Compression ratio R	5.5
Similarity parameter B	1.0342
Kinetic energy fraction T_0/E_0	0.2141
Gas energy fraction $U_{0,g}/E_0$	0.1477
Cosmic-ray energy fraction $U_{0,CR}/E_0$	0.6382

The calculations show that approximately 21.4% of the explosion energy remains in kinetic form, while only 14.8% is stored as thermal energy of the gas. In contrast, nearly 63.8% of the total energy is contained in the cosmic-ray component.

This result demonstrates the significant dynamical influence of cosmic rays in strongly modified shocks. The cosmic-ray pressure is not merely a small correction to the hydrodynamic solution; rather, it becomes the dominant energy reservoir within the remnant. Such behaviour is fundamentally different from the classical Sedov–Taylor solution, where the energy is primarily partitioned between kinetic motion and thermal gas.

The large cosmic-ray energy fraction obtained here supports the idea that supernova remnants can efficiently transfer explosion energy to relativistic particles. Consequently, cosmic rays play an important role in determining both the internal structure and long-term evolution of the remnant.

6 Conclusion

In this project, the structure of a cosmic-ray modified shock was investigated using the self-similar formulation developed by Chevalier. The dimensionless Euler equations describing the coupled evolution of thermal gas and cosmic-ray pressure were reduced to a system of nonlinear ordinary differential equations and solved numerically using the sixth-order Runge–Kutta method proposed by Luther (1968).

The numerical solution successfully reproduced the characteristic profiles of the dimensionless density, gas pressure, cosmic-ray pressure, and velocity throughout the supernova remnant. For the adopted parameters, $\beta = 0.5$, $\gamma_g = 5/3$, and $\gamma_{CR} = 4/3$, the effective adiabatic index was found to be $\gamma = 1.4444$, corresponding to a total compression ratio of $R = 5.5$. The similarity parameter was calculated to be $B = 1.0342$.

The energy analysis revealed that the largest fraction of the explosion energy is stored in the cosmic-ray component. Approximately 63.8% of the total energy resides in cosmic rays, compared to 21.4% in kinetic energy and 14.8% in thermal gas energy. These results demonstrate that cosmic rays can strongly influence the internal structure and energy distribution of a supernova remnant.

The present study highlights the importance of including cosmic-ray pressure in models of astrophysical shocks. Compared with the classical Sedov–Taylor solution, the modified shock solution predicts a substantially different energy partition, with relativistic particles becoming the dominant energy reservoir. This behaviour supports the widely accepted view that supernova remnants are efficient accelerators of Galactic cosmic rays and that cosmic-ray feedback must be considered in realistic descriptions of shock evolution.

A Numerical Implementation

A.1 Model Parameters

```
1 import numpy as np
2
3
4 beta = 0.5
5
6 gamma_g = 5.0/3.0
7 gamma_cr = 4.0/3.0
8
9 gamma = (5 + 3*(1-beta))/(3*(2-beta))
10
11 compression_ratio = (gamma + 1)/(gamma - 1)
12
13 A = (gamma + 1)/2
```

Listing 1: Physical parameters used in the modified shock calculations.

A.2 Implementation of the Governing Equations

```
1 def denominator(xi, F, U):
2
3     term1 = (gamma - 1)**2 * (
4         beta*gamma_g*F**(gamma_g-2)
5         + (1-beta)*gamma_cr*F**(gamma_cr-2)
6     )
7
8     term2 = 4*xi**2*(U - A*xi)**3
9
10    return term1 + term2
11
12
13 def dU_dxi(xi, F, U):
14
15     D = denominator(xi, F, U)
16
17     num1 = (gamma - 1)**2 * (
18         beta*F**(gamma_g-2)*(1.5*(gamma+1)*xi - 2*gamma_g*U)
19         +
20         (1-beta)*F**(gamma_cr-2)*(1.5*(gamma+1)*xi - 2*gamma_cr*U)
21     )
22
23     num2 = (
24         3*(gamma+1)*xi**3*U*(U - A*xi)**2
25     )
26
```

```

27     return (1/xi)*(num1/D + num2/D)
28
29
30 def dF_dxi(xi, F, U):
31
32     D = denominator(xi, F, U)
33
34     part1 = (
35         1.5*(gamma+1)*(gamma-1)**2 *
36         (
37             beta*F**(gamma_g-2)
38             +
39             (1-beta)*F**(gamma_cr-2)
40         )
41     )
42
43     part2 = (
44         xi*U*(8*U - (gamma+1)*xi)*(U - A*xi)**2
45     )
46
47     bracket = part1/D + part2/D
48
49     return -F/(U - A*xi)*bracket
50
51
52 def system(xi, y):
53
54     F = y[0]
55     U = y[1]
56
57     return np.array([
58         dF_dxi(xi, F, U),
59         dU_dxi(xi, F, U)
60     ])
61
62
63 def f(xi, y):
64     return system(xi, y)

```

Listing 2: Functions defining the dimensionless modified shock equations.

A.3 Luther Sixth-Order Runge–Kutta Integrator

```

1 sqrt21 = np.sqrt(21.0)
2
3 def rk6_step(x, y, h):
4
5     k1 = h*f(x, y)

```

```

6
7     k2 = h*f(
8         x + h,
9         y + k1
10    )
11
12    k3 = h*f(
13        x + h/2,
14        y + (3*k1 + k2)/8
15    )
16
17    k4 = h*f(
18        x + 2*h/3,
19        y + (8*k1 + 2*k2 + 8*k3)/27
20    )
21
22    k5 = h*f(
23        x + (7 - sqrt(21))*h/14,
24        y + (
25            3*(3*sqrt(21) - 7)*k1
26            - 8*(7 - sqrt(21))*k2
27            + 48*(7 - sqrt(21))*k3
28            - 3*(21 - sqrt(21))*k4
29        )/392
30    )
31
32    k6 = h*f(
33        x + (7 + sqrt(21))*h/14,
34        y + (
35            -5*(231 + 51*sqrt(21))*k1
36            -40*(7 + sqrt(21))*k2
37            -320*sqrt(21)*k3
38            +3*(21 + 121*sqrt(21))*k4
39            +392*(6 + sqrt(21))*k5
40        )/1960
41    )
42
43    k7 = h*f(
44        x + h,
45        y + (
46            15*(22 + 7*sqrt(21))*k1
47            +120*k2
48            +40*(7*sqrt(21) - 5)*k3
49            -63*(3*sqrt(21) - 2)*k4
50            -14*(49 + 9*sqrt(21))*k5
51            +70*(7 - sqrt(21))*k6
52        )/180
53    )

```

```

54
55     y_next = y + (
56         9*k1
57         +64*k3
58         +49*k5
59         +49*k6
60         +9*k7
61     )/180
62
63     return y_next

```

Listing 3: Implementation of the Luther (1968) sixth-order Runge–Kutta method.

A.4 Numerical Integration and Energy Calculations

```

1  h = -1e-4
2
3  xi_values = [1.0]
4  F_values = [1.0]
5  U_values = [1.0]
6
7  y = np.array([1.0, 1.0])
8  xi = 1.0
9
10 while xi > 0.01:
11
12     y = rk6_step(xi, y, h)
13
14     xi += h
15
16     xi_values.append(xi)
17
18     F_values.append(y[0])
19
20     U_values.append(y[1])
21
22 # Reconstruction of G and H
23
24 G_values = []
25 H_values = []
26
27 for xi, F, U in zip(xi_values, F_values, U_values):
28
29     rhs = (1 - gamma)/(2*xi**3)
30
31     factor = (U/xi - (gamma+1)/2)
32
33     G = rhs / (F**(1-gamma_g) * factor)

```



```

34     H = rhs / (F**(1-gamma_cr) * factor)
35
36     G_values.append(G)
37     H_values.append(H)
38
39 xi_values = np.array(xi_values)
40
41 F_values = np.array(F_values)
42
43 G_values = np.array(G_values)
44
45 H_values = np.array(H_values)
46
47 U_values = np.array(U_values)
48
49 # Energy integral
50
51 integrand = (
52     (1/(gamma-1))*F_values*U_values**2
53     +
54     (beta/(gamma_g-1))*G_values
55     +
56     ((1-beta)/(gamma_cr-1))*H_values
57 )
58
59 integrand = integrand * np.array(xi_values)**2
60
61 I = abs(np.trapezoid(integrand, xi_values))
62
63 B_minus5 = (
64     32*np.pi
65     /
66     (25*(gamma+1))
67 ) * I
68
69 B = B_minus5**(-1/5)
70
71 # Energy fractions
72
73 kinetic_integrand = (
74     (1/(gamma-1))
75     * np.array(F_values)
76     * np.array(U_values)**2
77     * np.array(xi_values)**2
78 )
79
80 gas_integrand = (
81     (beta/(gamma_g-1))

```

```

82     * np.array(G_values)
83     * np.array(xi_values)**2
84 )
85
86 cr_integrand = (
87     ((1-beta)/(gamma_cr-1))
88     * np.array(H_values)
89     * np.array(xi_values)**2
90 )
91
92 T = abs(np.trapezoid(kinetic_integrand, xi_values))
93
94 Ug = abs(np.trapezoid(gas_integrand, xi_values))
95
96 Ucr = abs(np.trapezoid(cr_integrand, xi_values))
97
98 total = T + Ug + Ucr
99
100 T_frac = T/total
101 Ug_frac = Ug/total
102 Ucr_frac = Ucr/total

```

Listing 4: Numerical integration, reconstruction of pressure profiles, and energy calculations.

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